

SAM TRITTO

DATA SCIENTIST

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EDUCATION

Master of Science in Analytics

Georgia Institute of Technology

Master of Education in Secondary Mathematics

Hawai'i Pacific University

Bachelor of Science in Mathematics

State University of New York

CORE SKILLS

Python, SQL, Data Science, Machine Learning, Statistics & Bayesian Statistics, Generative AI, Agentic AI, Google Cloud Platform, NLP, A/B Testing, Causal Inference, Tableau, Data Visualization, R, Probability, Streamlit, VertexAI & Kubernetes, MLOps, GitHub, Automation

PROFESSIONAL WORK EXPERIENCE

The Home Depot

Data Scientist

April 2024 - Present

• GenAI & Agentic AI

- **HR Document Automation:** Led the development of a GCP based AI workflow using Gemini models to parse 90k+ handwritten LOA documents annually utilizing the Salesforce API; **projected \$1.5M annual savings**.
- **VOA (Voice of the Associate) Survey Analysis:** Engineered a batch-processing AI template in GCP with Gemini models to classify and summarize over 1M survey responses in minutes; reclaiming **tens of thousands of leadership hours** biannually.
- **Graph RAG Agent:** Led the development a multi-modal and personalized Graph RAG SOP Agent (text/image) using Google's ADK for HR document retrieval and unstructured BigQuery insights. Built an Associate Knowledge Graph in BigQuery for personalization.
- **Associate and Location Embeddings & Semantic Layer:** Built and productionalized custom embeddings (AutoEncoder) for advanced AI reasoning as the foundation to the Semantic and Intelligence layers for our organization's AI initiatives.

• Machine Learning & Forecasting

- **Labor Supply Forecasting:** Developed and productionalized a 13-week forecast model to automate job requisitions recommendations; successfully piloted in stores driving an **estimated \$3M in annual cost avoidance**.
- **Optimization & Scale:** Reduced data footprint in my forecasting model by **98%** while still maintaining an **accuracy of 96%** with modern advanced models, BQML, and feature engineering.
- **Retention Modeling:** Developed a VertexAI Pipeline (Kubernetes) for imbalanced attrition classification; **saved \$1M in hiring costs** across 20 locations (**projected \$5M/year** nationwide).

• Leadership, Mentorship & Collaboration

- **Technical Upskilling:** Mentored the People Analytics team in Python and GenAI; developed workshop curricula to bridge the gap between raw data and business strategy; providing them with materials to self-learn and guides for best practices.
- **Technical Leadership:** Strengthened team technical maturity by mentoring analysts in robust model validation and the implementation of scalable, production-ready scripts; collaborating with IT to successfully deploy models across multiple business units.

Associate Data Scientist

April 2022 - April 2024

• Causal Inference & Experimentation

- **Experimental Design:** Implemented an ensemble of models (DiD, DoubleML, Causal Impact, Synthetic Controls, etc.) to quantify the impact and ROI of various pilot programs and experiments; built automated scripts to validate statistical assumptions.
- **Store Matching Tool:** Developed a clustering ensemble (K-Means, Gaussian Mixture, etc.), KNN, and Propensity Scores to help senior leaders decide which stores to use as pilot and control stores in their A/B test scenarios.

• Risk Assessment & Anomaly Detection

- **Unionization Risk Model:** Built a weighted ranking model utilizing web-scraped data (BLS, NLRB); optimized with Optuna to **increase accuracy by 400%**, currently used by VPs across multiple business units for store strategy.
- **Payroll Fraud Detection:** Created an outlier ensemble (Isolation Forest, Local Outlier Factor, OCSVM, AutoEncoder, and ECOD) for broad use currently being used to detect payroll anomalies, **saving an estimated \$10M annually**.

• Feature Engineering & Data Acquisition

- **Metrics Identification:** Developed a feature importance ensemble (Stepwise OLS Regression, LassoCV, Permutation Importance, Boruta SHAP, etc.) to identify and quantify measurable monthly proxy metrics for our "North Star" KPIs, focusing on associate engagement.
- **Automated Pipelines:** Developed API scripts to ingest multi-source data (NOAA, Yelp, US Census Bureau, etc.) for sentiment analysis and geographic risk modeling.

Hawaiian Airlines

Data Analyst (Sales Analytics)

June 2021 - April 2022

- **Data Engineering & ETL Automation**
 - **Pipeline Orchestration:** Architected end-to-end ETL pipelines using **Python** to automate the ingestion and transformation of disparate datasets, reducing manual reporting overhead by several hours weekly per stakeholder.
 - **Cloud Data Migration:** Participated in the strategic transition of legacy data architectures from Microsoft SQL Server to **Snowflake**, optimizing query performance and scalability for enterprise-level analytics.
 - **Infrastructure Optimization:** Developed robust scripts to query and validate large-scale data sources, ensuring high data integrity and “single source of truth” reliability across all downstream reporting.
- **Business Intelligence & Strategic Planning**
 - **Custom KPI Architecture:** Architected bespoke performance metrics with **Tableau**, translating complex business requirements into interactive visual narratives for executive decision-making.
 - **Cross Functional Analytics:** Collaborated with Point of Sale (POS) engineering team to deliver weekly deep-dive analyses, identifying operational bottlenecks and providing actionable insights to senior managers.
 - **Automated Insights Distribution:** Programmed automated report generation and distribution workflows, enabling real-time visibility into mission-critical KPIs for diverse business units.

Hawai'i Department of Education

Mathematics Teacher

April 2013 - September 2019

- **Strategic Leadership & Digital Transformation**
 - **Innovation Leadership & Technology Grant Acquisition:** Secured **\$90,000 in technology grants** via the Samsung Solve for Tomorrow STEM Competition; served as a primary spokesperson for digital initiatives, featuring in Fortune Magazine and NBC's Today Show with Al Roker to advocate for tech-driven solutions to local community problems.
 - **Digital Transformation & Engagement:** Spearheaded a department-wide digital transformation to re-engineer resource accessibility; leveraged engagement data to design interventions that mitigated chronic absenteeism and streamlined the information delivery.

AWARDS

Orange Spotlight Award

- Recognized by Executive Leadership for moving with speed, unlocking value, and team leadership. (The Home Depot, March 2026)

THD Hackathon 2026 - Third Place (Team Captain)

- Created an end to end and unbiased Agentic AI workflow for Talent Acquisition's hiring process using Google's ADK and Gemini, including a Workday mockup front end with Firebase Studio & Figma. AI suggested job descriptions, AI candidate resume parsing, AI unbiased candidate/job skills matching, AI interview panelist matching, AI automated interview scheduling using Microsoft's Graph API to block calendars, AI tailored interview questions based on our internal documents and candidate profile for a more interesting conversation, and an AI generated feedback letter for the candidate based on feedback from the Hiring Manager.

Kicking Up Dust Award

- Earned for applying AI to HR Document Automation from my Senior Director (The Home Depot, October 2025)

Fresh Paint Award

- Earned for applying AI to HR Document Automation from my VP (The Home Depot, November 2025)

Cybersecurity - Secure Squad

- Earned twice for placing in the top 4% for our Cybersecurity Team's phishing campaign (The Home Depot, 2023, 2025)

Homer Awards

- Continually recognized for living our company values; 4 Bronze, 3 Silver, 3 Gold, 3 Platinum
- Including an Executive Homer for participating in our Wage Investment project (The Home Depot, 2023)

Samsung Solve for Tomorrow STEM Competition

- Bamboo Bicycles - State Finalist - \$20,000 (Hawai'i Department of Education, 2014)
- Salt Water Powered Fans - State Winner - \$50,000 (Hawai'i Department of Education, 2016)
- Sustainable Taro Utensils - State Finalist - \$20,000 (Hawai'i Department of Education, 2019)

CERTIFICATIONS

Google Generative AI Leader Jan 2026 - Jan 2029

Tableau Data Analyst Jan 2021 - Jan 2022 (Expired)

MOOCs Over 150 courses completed, and still going!

SOFT SKILLS

Communication, Integrity, Humility, Critical Thinking, Creative Problem Solving, Emotional Intelligence, Leadership, Teamwork, Continual Learner, Technical Storytelling, Executive Communication, Thought Leadership